

Bayesian Classification

Advanced Statistical Inference

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Logistic Regression

1. Explain why a linear model $\mathbf{w}^\top \mathbf{x}$ is not sufficient by itself for binary classification if we want probabilistic predictions. Why does the sigmoid function solve this issue?
2. For logistic regression,

$$P(y = 1 \mid \mathbf{x}, \mathbf{w}) = \sigma(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x})}.$$

Given

$$\mathbf{w} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix},$$

compute $\mathbf{w}^\top \mathbf{x}$, $P(y = 1 \mid \mathbf{x}, \mathbf{w})$, and $P(y = 0 \mid \mathbf{x}, \mathbf{w})$.

3. Show that the Bernoulli likelihood for one labeled example can be written as

$$p(y_n \mid \mathbf{w}, \mathbf{x}_n) = \sigma(\mathbf{w}^\top \mathbf{x}_n)^{y_n} (1 - \sigma(\mathbf{w}^\top \mathbf{x}_n))^{1-y_n}.$$

4. Consider a dataset with two observations:

$$(\mathbf{x}_1, y_1) = ((1, 0)^\top, 1), \quad (\mathbf{x}_2, y_2) = ((1, 1)^\top, 0),$$

and parameter vector

$$\mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Compute:

- $\sigma(\mathbf{w}^\top \mathbf{x}_1)$ and $\sigma(\mathbf{w}^\top \mathbf{x}_2)$;
- the likelihood of each data point;
- the joint log-likelihood of the dataset.

Bayesian Logistic Regression

1. Assume the prior

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \mathbf{0}, \sigma_w^2 \mathbf{I}).$$

Write the posterior distribution

$$p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$$

up to proportionality. Why is this posterior not available in closed form?

2. Show that maximum a posteriori estimation for Bayesian logistic regression is equivalent to minimizing

$$\mathcal{U}(\mathbf{w}) = -\log p(\mathbf{y} \mid \mathbf{w}, \mathbf{X}) - \log p(\mathbf{w}).$$

What role does the prior play in this optimization problem?

3. Consider a one-dimensional logistic regression model with scalar parameter w , one observation $(x, y) = (2, 1)$, and prior

$$p(w) = \mathcal{N}(w \mid 0, 1).$$

Compute the log posterior up to an additive constant:

$$\log p(w \mid y, x) = \log p(y \mid w, x) + \log p(w) + \text{const.}$$

Evaluate this expression at $w = 0$ and $w = 1$ and say which value is larger.

4. Give one reason why each of the following can be used for Bayesian logistic regression, and one tradeoff:
 - MAP estimation;
 - Laplace approximation;
 - variational inference;
 - MM.

Prediction and Uncertainty

1. Suppose a Gaussian approximation to the posterior is

$$q(\mathbf{w}) = \mathcal{N}(\mathbf{w} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \boldsymbol{\mu} = \begin{pmatrix} 1 \\ -0.5 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 0.4 & 0 \\ 0 & 0.1 \end{pmatrix}.$$

For

$$\mathbf{x}_\star = \begin{pmatrix} 2 \\ 1 \end{pmatrix},$$

compute:

- the mean $\boldsymbol{\mu}^\top \mathbf{x}_\star$ of the latent score $f_\star$;
- the variance $\mathbf{x}_\star^\top \boldsymbol{\Sigma} \mathbf{x}_\star$;
- the approximate predictive probability using

$$\sigma \left(\frac{\boldsymbol{\mu}^\top \mathbf{x}_\star}{\sqrt{1 + \frac{\pi}{8} \mathbf{x}_\star^\top \boldsymbol{\Sigma} \mathbf{x}_\star}} \right).$$

2. In Bayesian classification, what is the difference between:

- a point prediction for the class label;
- a predictive probability;
- epistemic uncertainty;
- aleatoric uncertainty?

3. Suppose posterior samples give the following predictive probabilities for the positive class on a test point:

$$0.70, 0.80, 0.65, 0.75.$$

Use Monte Carlo integration to estimate $P(y_\star = 1 \mid \mathbf{x}_\star, \mathbf{y}, \mathbf{X})$. What does the spread of these four values suggest qualitatively about uncertainty?

Evaluation

1. A classifier on a binary test set yields the confusion matrix counts

$$\text{TP} = 18, \quad \text{TN} = 70, \quad \text{FP} = 6, \quad \text{FN} = 6.$$

Compute:

- accuracy;
- precision;
- recall;
- F1 score.

2. Explain why accuracy can be misleading on imbalanced datasets. Give a concrete example of a classifier with high accuracy but poor practical usefulness.

3. Suppose we have three posterior samples for the probability of the true test label:

$$p(y_\star \mid \mathbf{w}^{(1)}, \mathbf{x}_\star) = 0.9, \quad p(y_\star \mid \mathbf{w}^{(2)}, \mathbf{x}_\star) = 0.6, \quad p(y_\star \mid \mathbf{w}^{(3)}, \mathbf{x}_\star) = 0.3.$$

Compute the Monte Carlo approximation of the predictive test log-likelihood

$$\log p(y_\star \mid \mathbf{x}_\star, \mathbf{y}, \mathbf{X}) \approx \log \left(\frac{1}{3} \sum_{s=1}^3 p(y_\star \mid \mathbf{w}^{(s)}, \mathbf{x}_\star) \right).$$

4. What does it mean for a classifier to be calibrated? Briefly explain the idea behind a reliability diagram and the expected calibration error.